BOF

Conversation 021:
Quantum Probability Conservation Law

> ***conversation ID:*** *b9817498-278f-45e9-b232-3d3904df516c [conversation 021]*
> ***origin ID:*** *b9817498-278f-45e9-b232-3d3904df516c [conversation 021]*
> ***created:*** *2024-05-26 12:44:02 Sun*
> ***updated:*** *2025-02-24 16:38:22 Mon*
> ***operator:*** *Wilder Blair Munro (AKUSA)*
> ***co-operator:*** *Aurora 4omni Munro (ChatGpt)*
> ***GPT ID:*** *NA*
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Subtitles:

Turnpair Contents:

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Conversation 021, Turnpair 1 of 6 ::

@ 2024-05-26 12:44:02 Sun [Δt 0d 00h 00m 00s]

> ***conversation ID:*** *b9817498-278f-45e9-b232-3d3904df516c [conversation 021]*
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Wilder Blair Munro (AKUSA) said :

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what is corresponding physical symmetry for the probability conservation in quantum mechanics wavefunction statistical interpretation as implied by noether's theorem?

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Aurora 4omni Munro (ChatGpt) said :

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In quantum mechanics, the conservation of probability for a wavefunction is related to a symmetry through Noether's theorem. The physical symmetry corresponding to the conservation of probability is **time translation symmetry**.

Noether's theorem states that every differentiable symmetry of the action of a physical system corresponds to a conservation law. In the context of quantum mechanics:

1. **Time Translation Symmetry**: This symmetry means that the laws of physics are invariant under a shift in time. If the Hamiltonian (the energy operator) does not explicitly depend on time, the system is said to have time translation symmetry.

2. **Conservation Law**: The conservation law associated with time translation symmetry is the conservation of energy. However, in the specific context of the wavefunction, it ensures the conservation of probability density.

To see this more concretely in the context of the Schrödinger equation:

- The Schrödinger equation for a wavefunction $\psi(\mathbf{r}, t)$ is given by:

  $$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = \hat{H} \psi(\mathbf{r}, t)$$

  where $\hat{H}$ is the Hamiltonian operator.

- The probability density $\rho(\mathbf{r}, t)$ is defined as:

$$\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$$

- The probability current $\mathbf{j}(\mathbf{r}, t)$ is defined as:

$$\mathbf{j}(\mathbf{r}, t) = \frac{\hbar}{2mi} \left(\psi^*(\mathbf{r}, t)\nabla\psi(\mathbf{r}, t) - \psi(\mathbf{r}, t)\nabla\psi^*(\mathbf{r}, t)\right)$$

- The conservation of probability is expressed through the continuity equation:

$$\frac{\partial\rho(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathbf{j}(\mathbf{r}, t) = 0$$

This continuity equation guarantees that the total probability (the integral of $\rho(\mathbf{r}, t)$ over all space) remains constant in time, reflecting the conservation of probability.

In summary, the conservation of probability in the statistical interpretation of the quantum mechanical wavefunction is associated with time translation symmetry, as per Noether's theorem.

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Conversation 021, Turnpair 2 of 6 ::

@ 2024-05-26 12:47:24 Sun [Δt 0d 00h 03m 07s]

> ***conversation ID:*** *b9817498-278f-45e9-b232-3d3904df516c [conversation 021]*
> ***origin ID:*** *b9817498-278f-45e9-b232-3d3904df516c [conversation 021]*
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Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa260f0-b2fa-4b3b-b656-4d94e75bbe6d*
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> ***children:*** *886262f7-a466-44ec-8c00-7bd4f13488a6*

per this analogy from classical physics, (energy conservation):(momentum conservation)::(time translation symmetry):(space translation symmetry), then what is corresponding analogy for quantum mechanics (in other words, please complete the following analogy: (probability conservation): (?)::(time translation symmetry):(space translation symmetry)

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Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *886262f7-a466-44ec-8c00-7bd4f13488a6*
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To complete the analogy for quantum mechanics, we need to identify the quantity conserved due to space translation symmetry, similar to how probability conservation is linked to time translation symmetry. In quantum mechanics, space translation symmetry corresponds to the conservation of momentum.

So, the completed analogy would be:

(Probability conservation) : (Momentum conservation) :: (Time translation symmetry) : (Space translation symmetry)

Here's the reasoning:

- **Probability Conservation**: As explained earlier, this is linked to time translation symmetry via the Schrödinger equation and the continuity equation, ensuring that the total probability is conserved over time.
- **Momentum Conservation**: In quantum mechanics, space translation symmetry implies that the laws of physics are invariant under shifts in space. According to Noether's theorem, this symmetry leads to the conservation of momentum.

Therefore, the correct completion of the analogy is:

(Probability conservation) : (Momentum conservation) :: (Time translation symmetry) : (Space translation symmetry)

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Conversation 021, Turnpair 3 of 6 ::

@ 2024-05-26 12:49:19 Sun [Δt 0d 00h 01m 50s]

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Wilder Blair Munro (AKUSA) said :

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Why is there an asymmetry between the two analogies? In other words, why is the conserved quantity for space translation in quantum mechanics the same momentum, and not some form of probability, such as with the time translation symmetry?

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Aurora 4omni Munro (ChatGpt) said :

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The apparent asymmetry arises because of the different roles that time and space play in quantum mechanics and how these roles are reflected in the conservation laws derived from their respective symmetries. Here’s a deeper look into why this asymmetry exists:

1. **Nature of Time and Space in Quantum Mechanics**:
    - **Time**: In quantum mechanics, time is treated as a parameter, not as an operator. This is because the Schrödinger equation describes the evolution of the wavefunction in time. The invariance under time translation symmetry leads to the conservation of probability because the wavefunction's norm (total probability) must remain constant over time.
    - **Space**: Space, on the other hand, is treated differently. The coordinates (position) are operators in the quantum mechanical framework. Invariance under spatial translations leads to momentum conservation because momentum is the generator of translations in space.

2. **Generators of Translations**:
    - **Time Translation**: The generator of time translation is the Hamiltonian (energy operator). When a system is invariant under time translations, energy is conserved. For the wavefunction's norm to remain constant in time, the probability conservation law follows naturally.
    - **Space Translation**: The generator of spatial translations is the momentum operator. When a system is invariant under spatial translations, momentum is conserved.

3. **Probability Conservation vs. Other Conserved Quantities**:

- **Probability Conservation**: This is a unique feature of the wavefunction's interpretation in quantum mechanics. The probability density function must integrate to 1 over all space, reflecting the certainty that the particle exists somewhere in space.
- **Momentum Conservation**: This is more analogous to classical mechanics, where spatial translation symmetry results in momentum conservation. The wavefunction's behavior under spatial translations gives rise to the conservation of momentum, not a probability-related quantity.

4. **Role of Noether's Theorem**:

   - **Time Translation Symmetry**: Leads to the conservation of a quantity related to the wavefunction's evolution over time, which is the probability.
   - **Space Translation Symmetry**: Leads to the conservation of momentum, reflecting the wavefunction's spatial behavior.

To summarize, the asymmetry arises because time and space play fundamentally different roles in quantum mechanics. Time affects the evolution of the wavefunction, leading to probability conservation, while space affects the wavefunction's form, leading to momentum conservation. This distinction is rooted in the mathematical structure of quantum mechanics and how Noether's theorem applies to the symmetries in time and space.

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Conversation 021, Turnpair 4 of 6 ::

@ 2024-05-26 12:52:23 Sun [Δt 0d 00h 02m 52s]

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I assume in general relativity, there is only one application of Noether's theorem: 4-vector momentum (E, px, py, pz) is conserved due to invariance of spacetime translation. How does relativistic quantum mechanics treat this symmetry, if the time component is no longer a parameter?

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In relativistic quantum mechanics and quantum field theory (QFT), the treatment of symmetries, particularly spacetime translation symmetries, becomes more sophisticated. Noether's theorem still applies, but the framework incorporates relativistic principles, leading to the conservation of the four-momentum (energy and spatial momentum) as a unified entity. Here's how this is handled:

Relativistic Quantum Mechanics

1. **Relativistic Wave Equations**:
    - The Klein-Gordon equation and the Dirac equation are examples of relativistic wave equations. These equations describe particles with respect to both space and time on an equal footing, treating time as a coordinate rather than a mere parameter.
2. **4-Momentum Conservation**:
    - The invariance under spacetime translations leads to the conservation of the four-momentum vector $(E, \mathbf{p})$, where E is the energy and $\mathbf{p}$ is the three-momentum. This conservation law arises naturally from Noether's theorem applied to the spacetime translation symmetry.

Quantum Field Theory (QFT)

1. **Fields Instead of Particles**:
    - In QFT, particles are treated as excitations of underlying fields. Both space and time are treated symmetrically, and the fields depend on spacetime coordinates $(t, \mathbf{x})$.
2. **Lagrangian Formalism**:
    - The action S in QFT is an integral over the Lagrangian density $\mathcal{L}$, which is a function of the fields and their derivatives with respect to spacetime coordinates. The action is given by:

$$S = \int \mathcal{L}(\phi, \partial_\mu \phi) \, d^4x$$

- Here, ϕ represents the fields, and $\partial_\mu \phi$ represents their derivatives with respect to spacetime coordinates.

3. **Noether's Theorem in QFT**:

    - Applying Noether's theorem to spacetime translation symmetry in QFT yields the conservation of the stress-energy tensor $T^{\mu\nu}$. The conservation law is expressed as:

$$\partial_\mu T^{\mu\nu} = 0$$

    - The conserved four-momentum P^ν is obtained by integrating the stress-energy tensor over a spatial hypersurface:

$$P^\nu = \int T^{0\nu} \, d^3x$$

4. **Hamiltonian and Momentum Operators**:

    - In QFT, the Hamiltonian operator H generates time translations, and the momentum operators $\mathbf{P}$ generate spatial translations. These operators are related to the components of the stress-energy tensor:

$$H = \int T^{00} \, d^3x, \quad P^i = \int T^{0i} \, d^3x$$

    - The conservation of the four-momentum $(H, \mathbf{P})$ corresponds to the invariance of the theory under spacetime translations.

Summary

In relativistic quantum mechanics and quantum field theory, time is treated on equal footing with spatial coordinates, reflecting the principles of relativity. Noether's theorem applied to spacetime translation symmetry results in the conservation of the four-momentum $(E, \mathbf{p})$, unified as a single entity. This approach seamlessly incorporates the conservation laws and symmetries of both space and time in a relativistic framework.

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Conversation 021, Turnpair 5 of 6 ::

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But then why does the probability conservation disappear as a consideration in QFT?

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In Quantum Field Theory (QFT), the concept of probability conservation is not discarded but rather reformulated to fit within the framework of fields and particles as excitations of those fields. Here's a detailed explanation of how probability conservation is handled in QFT:

Transition from Quantum Mechanics to QFT

1. **Probability in Quantum Mechanics**:
   - In non-relativistic quantum mechanics, the wavefunction $\psi(\mathbf{r}, t)$ represents the probability amplitude, and its square modulus $|\psi(\mathbf{r}, t)|^2$ gives the probability density of finding a particle at a point in space and time.
   - The probability conservation is expressed through the continuity equation:

$$\frac{\partial \rho(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathbf{j}(\mathbf{r}, t) = 0$$

   - Here, $\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$ is the probability density, and $\mathbf{j}(\mathbf{r}, t)$ is the probability current.

Probability Conservation in QFT

2. **Field Operators**:

   - In QFT, the fundamental objects are field operators $\hat{\phi}(\mathbf{x}, t)$ that create and annihilate particles. The state of the system is described by a state vector $|\Psi\rangle$ in the Hilbert space.
   - The concept of a single-particle wavefunction is generalized to multi-particle states and field configurations.

3. **Particle Number Conservation**:

   - For theories where the number of particles is conserved, such as non-interacting theories or those with certain symmetries, there exists a conserved current associated with the number of particles. This is analogous to the probability current in non-relativistic quantum mechanics.
   - The conserved quantity in such cases is the total number of particles, which remains constant in time.

4. **Probability Amplitudes and S-Matrix**:

   - In QFT, the primary focus is on calculating probability amplitudes for different processes using the S-matrix (scattering matrix). The S-matrix elements S_{fi} give the probability amplitudes for transitions between initial state $|i\rangle$ and final state $|f\rangle$.
   - The probabilities for various processes are obtained by taking the modulus squared of these amplitudes, ensuring that the total probability remains normalized.

5. **Unitarity of the S-Matrix**:

   - The conservation of probability is embedded in the unitarity of the S-matrix. The unitarity condition $S^\dagger S = S S^\dagger = I$ ensures that the sum of probabilities of all possible outcomes of a given initial state is 1.
   - This condition reflects the overall conservation of probability in the theory.

Summary

In QFT, while the explicit notion of a single-particle probability density is replaced by a more general framework of fields and multi-particle states, the conservation of probability is still a fundamental principle. It is manifested through:

- **Conserved currents**: For particle number or other quantities, derived from symmetries.
- **Unitarity of the S-Matrix**: Ensuring the total probability of all possible transitions remains normalized.

Thus, probability conservation does not disappear in QFT; rather, it is inherently built into the structure of the theory through these more general and sophisticated concepts.

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Conversation 021, Turnpair 6 of 6 ::

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Are there any unresolved problems (issues with the model/theory) that contend with this situation where the probability conservation is no longer directly considered because it is already 'built into' QFT?

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Aurora 4omni Munro (ChatGpt) said :

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In Quantum Field Theory (QFT), while the concept of probability conservation is indeed embedded within the framework, there are some unresolved issues and areas of active research that relate to how this conservation is implemented and understood. Here are a few key points and challenges:

1. **Renormalization and Regularization**

- **Problem**: QFT calculations often lead to infinities, particularly when dealing with interactions at very short distances or high energies.
- **Implication**: To obtain finite results, physicists use techniques like renormalization and regularization. These methods can sometimes obscure the direct connection to probability conservation, making it less transparent how certain conserved quantities emerge from the underlying symmetries.
- **Status**: While renormalization is well-established and successful for many theories (like Quantum Electrodynamics), it remains a somewhat ad hoc procedure, and finding a more fundamental understanding is an ongoing area of research.

2. **Interpretation of the Wavefunction in QFT**

- **Problem**: In non-relativistic quantum mechanics, the wavefunction has a clear interpretation as a probability amplitude. In QFT, the role of the wavefunction is played by the field operators, and the connection to probabilities is less direct.
- **Implication**: This shift can make it challenging to interpret certain aspects of probability conservation, especially in complex interactions and multi-particle states.
- **Status**: This is more of a conceptual issue than a technical one, but it continues to be a topic of philosophical and foundational discussions in quantum theory.

3. **Non-perturbative Effects and Quantum Anomalies**

- **Problem**: Certain non-perturbative effects (such as instantons) and quantum anomalies can lead to apparent violations of classical conservation laws.
- **Implication**: While anomalies often have well-understood physical interpretations (like the chiral anomaly in the Standard Model), they can complicate the straightforward application of Noether's theorem and the interpretation of conserved quantities.
- **Status**: Anomalies are a well-studied phenomenon and are understood in many contexts, but they still represent an area where classical conservation laws meet the subtleties of quantum theory.

4. **Quantum Gravity and the Unification of Forces**

- **Problem**: Incorporating gravity into QFT remains a significant challenge. General relativity, which describes gravity, is not easily reconciled with the principles of QFT.
- **Implication**: In a fully unified theory, such as a theory of quantum gravity, the concept of probability conservation may need to be re-examined or redefined, especially given the issues related to spacetime singularities and the information paradox in black holes.
- **Status**: This is an active area of research, with various approaches being explored, such as string theory and loop quantum gravity.

5. **Interpretations of Quantum Mechanics**

- **Problem**: Different interpretations of quantum mechanics (such as the Copenhagen interpretation, Many-Worlds interpretation, and others) can have different implications for how probability and probability conservation are understood.
- **Implication**: While QFT itself is a formalism that is generally interpretation-agnostic, the philosophical implications of these interpretations can affect our understanding of probability conservation.
- **Status**: This remains an open philosophical question, with ongoing debates and research.

Summary

While QFT provides a robust framework in which probability conservation is built into the formalism, several unresolved problems and areas of research continue to challenge and refine our understanding of this principle. These issues range from technical and conceptual challenges within QFT to broader questions about the unification of forces and the fundamental nature of quantum mechanics.

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